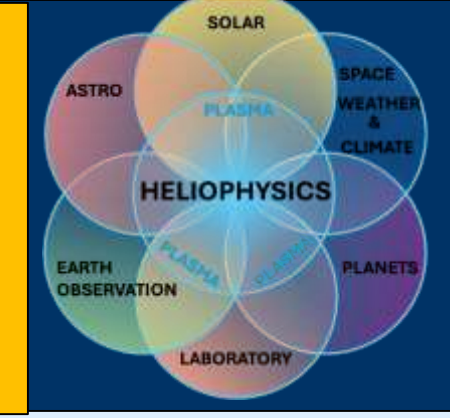




SolarShield: Physics-Informed Machine Learning for Multi-Horizon LEO Altitude-Change Forecasting Under Space-Weather Variability

Malhar Gajjar

B.Sc. Data Science, School of Technology, Pandit Deendayal Energy University, Gandhinagar, India



Brief Motivation

- Low-Earth-orbit (LEO) satellites experience atmospheric drag that varies strongly with thermospheric density and relative velocity. Solar and geomagnetic disturbances can rapidly alter the upper atmosphere, increasing uncertainty in short-term orbit evolution.
- Conventional propagation alone may not fully capture these nonlinear, time-varying effects. SolarShield combines space-weather observations, orbital dynamics and physics-derived atmospheric features with machine learning to forecast LEO altitude changes at multiple lead times.

Objective

- Develop a physics-aware multi-horizon forecasting framework for predicting LEO altitude changes at 1 h, 6 h, 15 h, and 24 h using coupled space-weather, thermospheric, and orbital information.
- Capture the effect of space-weather-driven atmospheric drag on short-term LEO orbital response using physically derived features.
- Evaluate forecast reliability across nominal and disturbed conditions by comparing physics-aware machine-learning and deep-learning models against baseline approaches using central-error, tail-error, bias, and drag-regime diagnostics.

Methodology

Physics-Aware SolarShield Forecasting Pipeline :

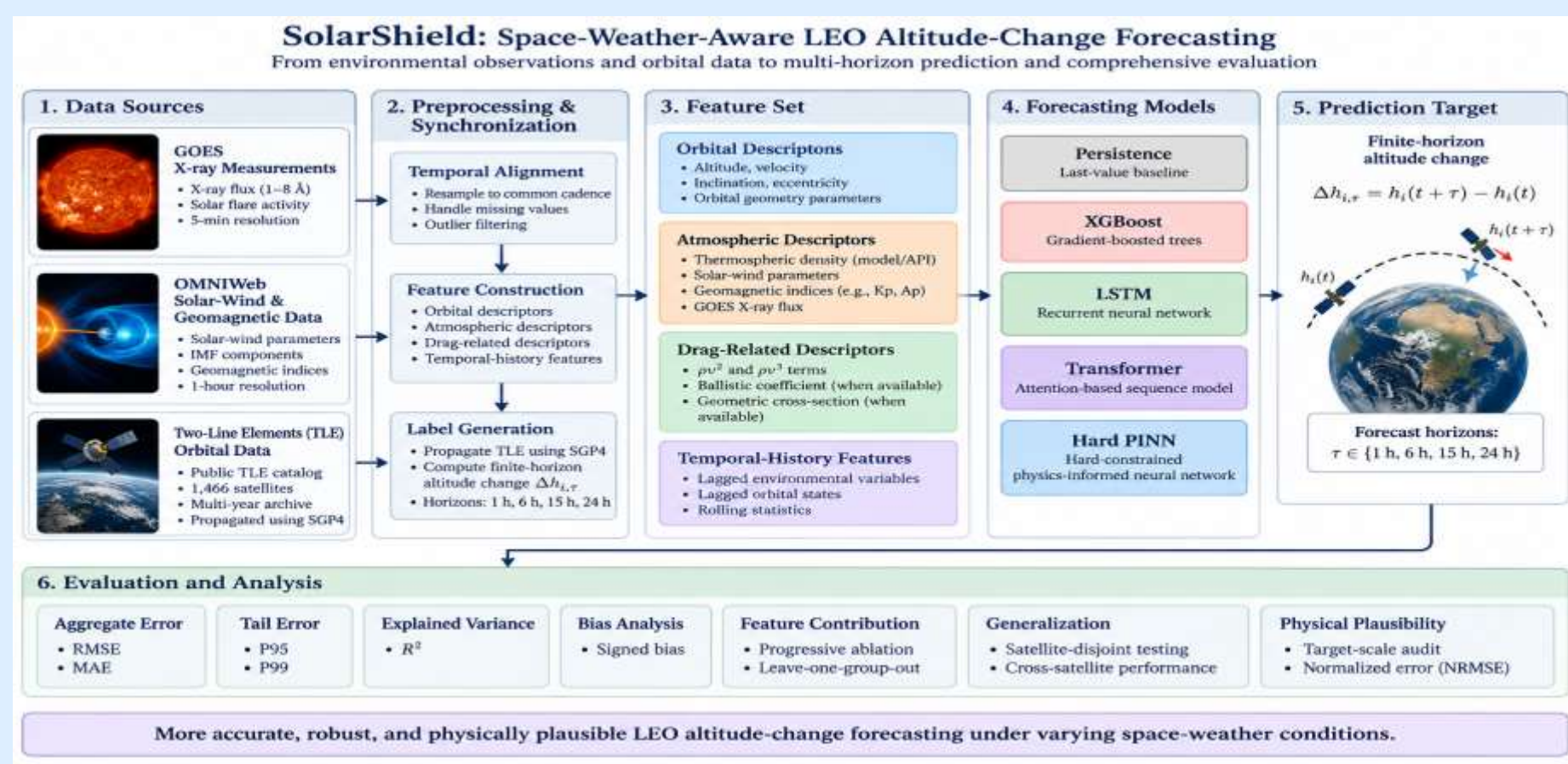


Figure 1. End-to-end SolarShield methodology from space-weather inputs to multi-horizon altitude forecasts.

- Performance & Physical Diagnostics: Forecasts were evaluated using RMSE, MAE, Bias, (R^2), P95, and P99, together with error analysis across space-weather activity and atmospheric-drag regimes.

$$P_{\rho} = \rho v_{\text{rel}}^3, D_{\rho} = \rho v_{\text{rel}}^2$$

- Data Integration: Hourly GOES XRS-B, OMNI solar-wind/geomagnetic parameters, and TLE-derived SGP4 orbital states were synchronized for 1,466 LEO/near-LEO satellites (2020–2024) using a 24 h historical input window.
- Physics-Aware Feature Engineering: Thermospheric density from an MSIS atmospheric model was combined with orbital quantities to construct drag-sensitive features such as: $a \cdot D = -\frac{1}{2} C_D \frac{A}{m} \rho v_{\text{rel}}^2$ and $\epsilon = -\frac{\mu}{2a}$ with inputs including (ρ) , (v_{rel}) , (ρv_{rel}^2) , (ρv_{rel}^3) , semi-major axis, orbital energy, and lagged/rolling space-weather variables.
- The 24 h physics-aware feature sequence was used to predict LEO altitude change at $[\tau = 1, 6, 15, 24 \text{ h}]$ using Persistence, XGBoost, LSTM, Transformer, soft PINN, and hard drag-PINN models.
- Performance & Physical Diagnostics: Models were evaluated using RMSE, MAE, Bias, (R^2), P95, and P99, together with diagnostics across space-weather activity and atmospheric-drag regimes to assess forecast reliability under both nominal and disturbed conditions.

Results & Physical Interpretation

Solution:

- Physics-aware XGBoost achieved the strongest overall forecasting performance, with test RMSE of 23.92 km (1 h), 50.29 km (6 h), and 61.63 km (15 h), while forecast uncertainty increased systematically with prediction horizon.
- Forecast performance was strongly evaluated across atmospheric-drag regimes, using the physics-based forcing proxy

$$[D_{\rho} = \rho v_{\text{rel}}^2]$$

to assess model behaviour from quiet to enhanced thermospheric-drag conditions.

- Tail-error and bias diagnostics revealed forecast reliability beyond average accuracy, using MAE, Bias, (R^2), P95 and P99 to identify rare but operationally important high-error conditions:

$$[Bias = \frac{1}{N} \sum_{i=1}^N (\widehat{\Delta h}_i - \Delta h_i)]$$

- Physics-aware features improved the interpretability of orbital-response forecasts by linking thermospheric density and drag forcing directly to altitude-change predictions.

Results:

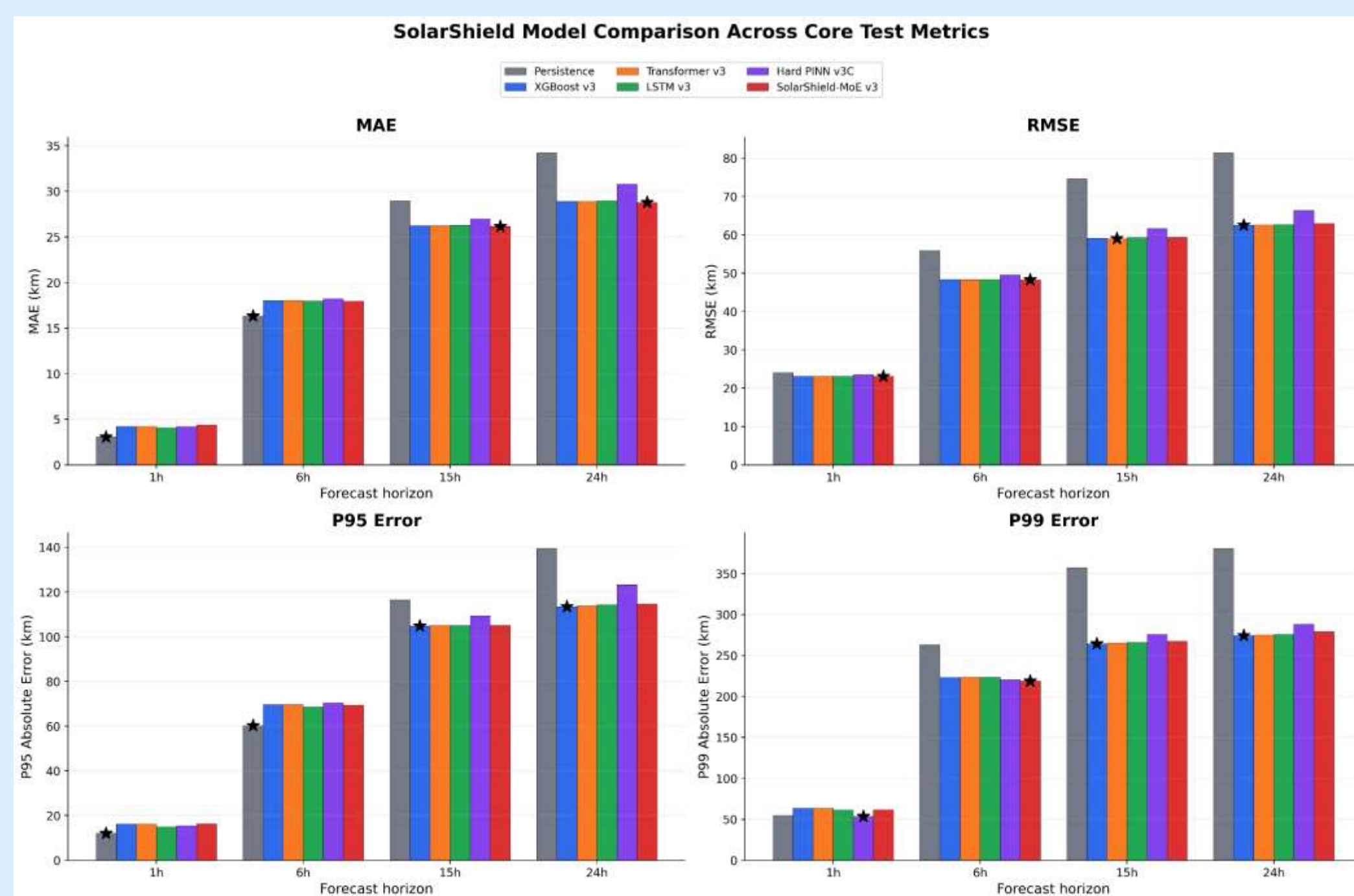


Figure 2. Model performance across forecast horizons using MAE, RMSE, P95, and P99 errors. Lower is better; ★ denotes the best-performing model at each horizon.

Key Findings & Operational Relevance

- Physics-aware features connect space-weather forcing, thermospheric density, atmospheric drag, and LEO orbital response within a unified forecasting framework.
- XGBoost achieved the strongest overall non-ensemble performance, demonstrating the value of physically informed features for short-term orbit forecasting.
- Forecast uncertainty increased with prediction horizon, highlighting the need for horizon-specific reliability assessment.
- P95/P99 and drag-regime diagnostics exposed high-error conditions that average RMSE alone can obscure.
- SolarShield offers a pathway toward storm-time orbit monitoring, conjunction-analysis support, and flight-dynamics decision assistance.

Bibliography/ References

- J. M. Picone *et al.*, “NRLMSISE-00 empirical model of the atmosphere,” *JGR: Space Physics*, 107(A12), 2002.
- D. A. Vallado *et al.*, “Revisiting Spacetrack Report #3,” *AIAA/AAS Astrodynamics Specialist Conf.*, 2006.
- G. Acciarini *et al.*, “Improving Thermospheric Density Predictions in Low-Earth Orbit With Machine Learning,” *Space Weather*, 22, 2024.
- H. Xu *et al.*, “A VMD-SVM Method for LEO Satellite Orbit Prediction with Space Weather Parameters,” *Remote Sensing*, 17, 746, 2025.
- M. S. Mahmud *et al.*, “Physics-informed neural network for predicting orbital parameters of Low Earth Orbit satellites using two line element dataset,” *Adv. Space Res.*, 77, 2026.
- E. Camporeale, “The Challenge of Machine Learning in Space Weather: Nowcasting and Forecasting,” *Space Weather*, 17, 2019.
- R. J. Licata *et al.*, “Machine-Learned HASDM Thermospheric Mass Density Model With Uncertainty Quantification,” *Space Weather*, 20, 2022.
- D. M. Oliveira *et al.*, “Modeling Thermosphere Density Enhancements During Extreme Magnetic Storms,” *Front. Astron. Space Sci.*, 8, 2021.